		the variation of the tension = difference in the values of $T = 2mg$
11	**B**	Option A : All geostationary satellites are of same distance of the Earth, hence, they experience the same gravitational field strength. Since the masses of satellites are different, $F = mg$, they will experience different amount of force. (Correct statement) Option B : Since they experience the same gravitational field strength, they will experience the same centripetal acceleration. (Incorrect statement) Option C : All geostationary satellites of Earth have a period of 24 hrs. Hence, they have the same angular velocities. (Correct statement) Option D : $v = r\omega$. All geostationary satellites have the same linear speed as they have the same angular velocities. However, because the satellites have different mass, they will have different kinetic energies. (Correct statement)
12	**C**	The frequency response curve of a damped oscillation should be entirely below (within) the